

```
1  #include "AutoCaesarIterator.h"
2
3  AutoCaesarIterator::AutoCaesarIterator(const char aInit[KEY_LENGTH],
4      const std::string* aCollection) noexcept :
5      fCollection(aCollection),
6      fKeys(aInit),
7      fInit(aInit),
8      fIndex(0)
9  {
10
11  }
12
13  AutoCaesarIterator::value_type AutoCaesarIterator::operator*() const      ↗
14      noexcept
15  {
16      char lResult = fInit[fIndex];
17
18      if (std::isalpha(lResult))
19      {
20          bool lIsLower = std::islower(lResult) != 0;
21          lResult = std::toupper(lResult);
22          lResult = (char)*fKeys;
23
24          if (lResult > 'Z')
25          {
26              lResult -= 26;
27          }
28
29          if (lIsLower != 0)
30          {
31              lResult = std::tolower(lResult);
32          }
33      }
34
35      return lResult;
36  }
37
38  AutoCaesarIterator::iterator& AutoCaesarIterator::operator++() noexcept
39  {
40
41      char lCurrent = (*this).operator*();
42      if (std::isalpha(lCurrent))
43      {
44          fKeys += lCurrent;
45          fKeys++;
46      }
47
48      fIndex++;
```

```
49     return *this;
50
51
52 }
53 AutoCaesarIterator::iterator AutoCaesarIterator::operator++(int) noexcept
54 {
55     iterator lCurrent= *this;
56     (*this)++;
57     return lCurrent;
58 }
59
60 bool AutoCaesarIterator::operator==(const iterator& aOther) const noexcept
61 {
62     return fCollection == aOther.fCollection &&
63            fInit == aOther.fInit &&
64            fIndex == aOther.fIndex;
65 }
66
67 AutoCaesarIterator::iterator AutoCaesarIterator::begin() const noexcept
68 {
69     iterator lBegin = *this;
70     lBegin.fKeys = KeyIterator<KEY_LENGTH>(fInit);
71     lBegin.fIndex = 0;
72     return lBegin;
73
74
75 }
76 AutoCaesarIterator::iterator AutoCaesarIterator::end() const noexcept
77 {
78     iterator lEnd = *this;
79     lEnd.fIndex = fCollection->size();
80     return lEnd;
81
82 }
83
84
85
```